Logistic Regression

CSE 329

Given ��, we approximate �� with a linear function.

�� is called parameters/weights

$$h(x) = \sum_{i=0}^{d} \theta_i x_i = \theta^T x$$

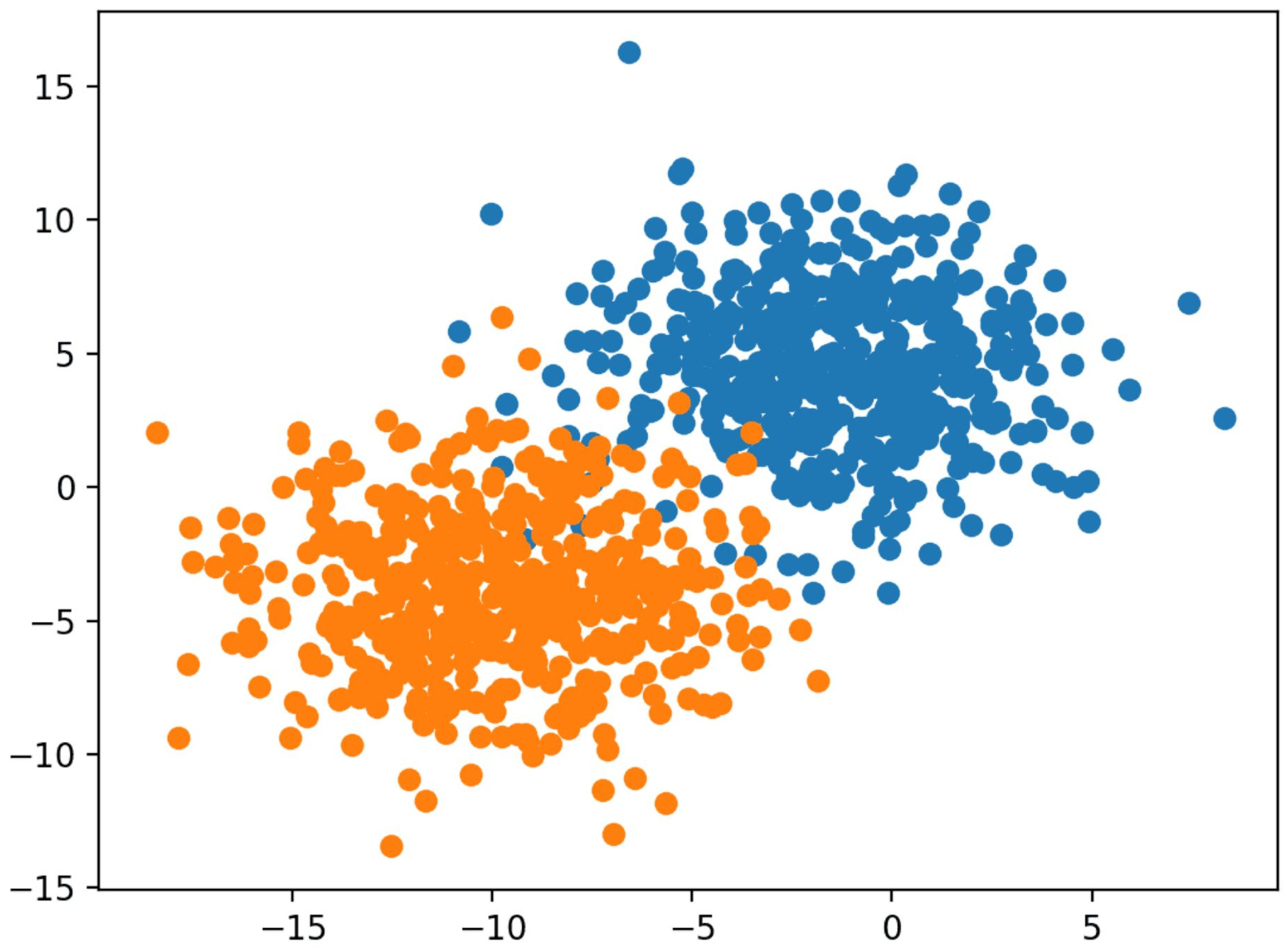
What about classification?

Y is a discrete target space

Commonly binary classification tasks

��∈{0, 1}

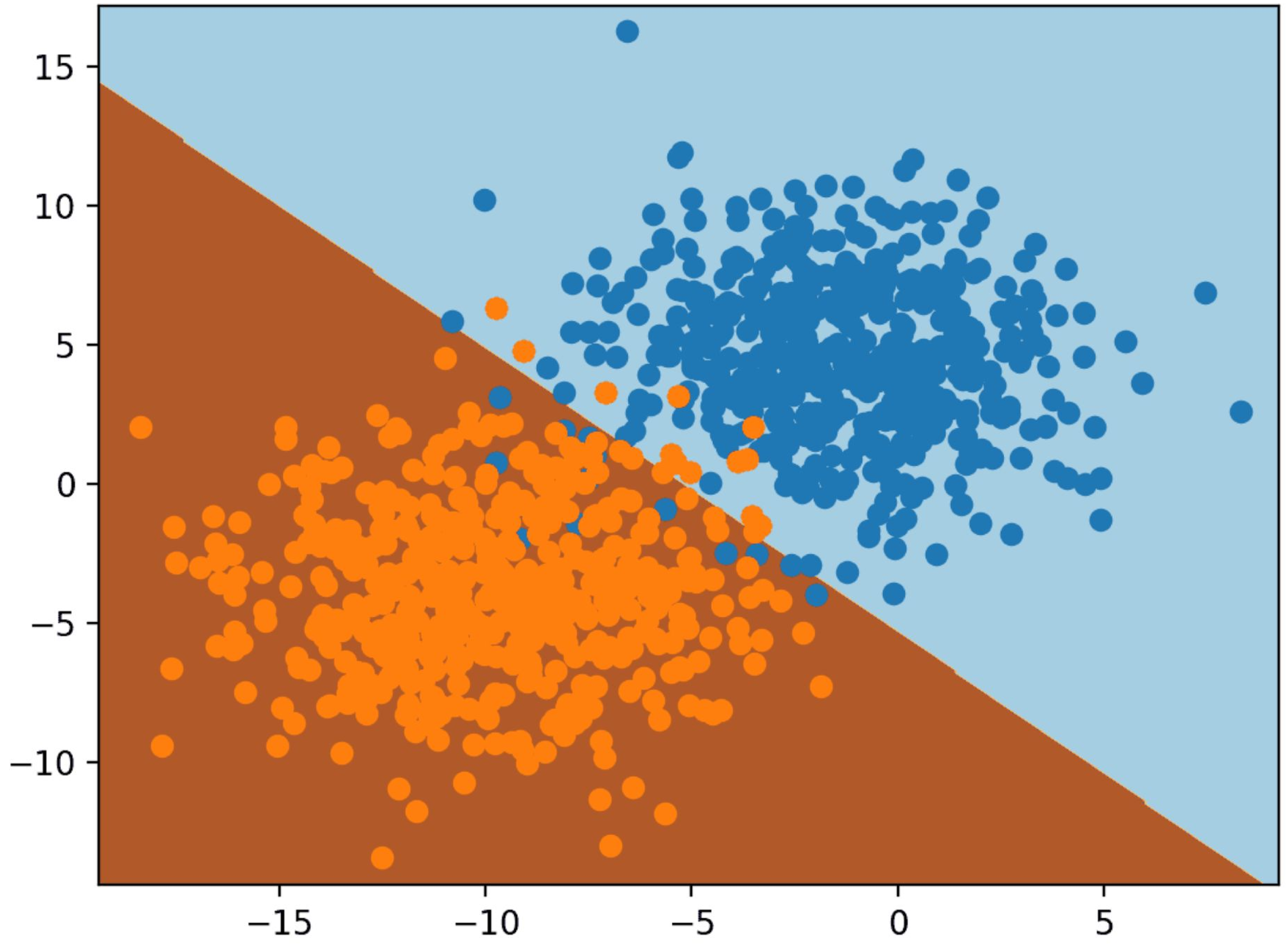
Often called positive class vs negative class



Instead of predicting ��∈0, 1, we can predict ℙ(��=1)

$$h_\theta(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}},$$

$$g(z) = \frac{1}{1 + e^{-z}}$$



Why this function?

	To Predict	Range		
		Min	Middle	Max
Binary Classification	y	[illegible]	-	[illegible]
Predict Probability	�� = ����=1, �� =1−��	[illegible]	[illegible].[illegible]	[illegible]
Odds	�� 1−	[illegible]	[illegible]	[illegible]
Log-odds	log��1− ��	−[illegible]	[illegible]	[illegible]
Linear Regression Prediction		−[illegible]	[illegible]	[illegible]

$$h_\theta(x) = \theta^T x = \log\left(\frac{p}{1-p}\right)$$

$$e^{\theta^T x} = \frac{p}{1-p}$$

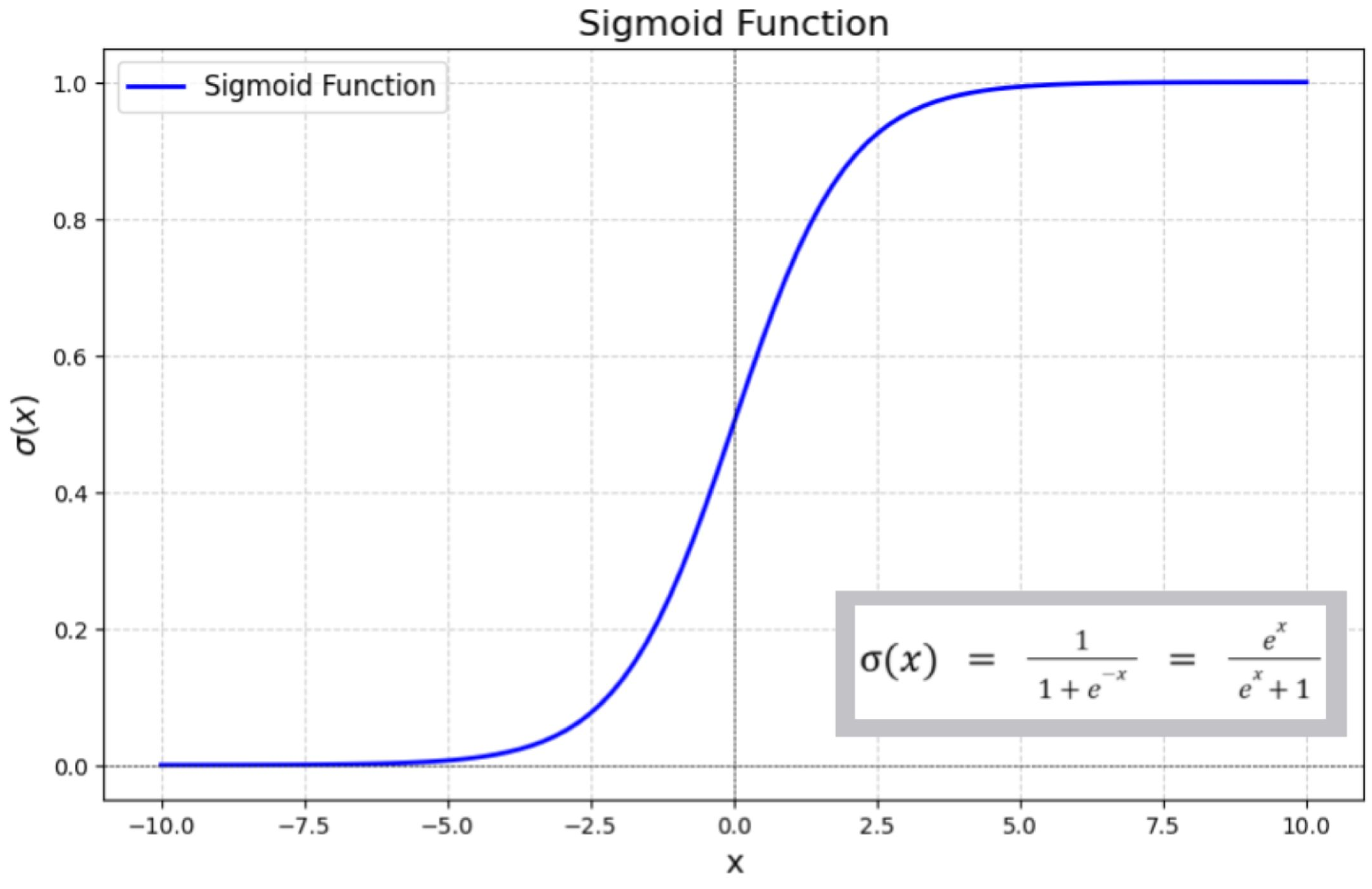
$$e^{\theta^T x} - e^{\theta^T x} p = p$$

$$e^{\theta^T x} = p(1 + e^{\theta^T x})$$

$$p = \frac{e^{\theta^T x}}{1 + e^{\theta^T x}}$$

$$p = \frac{1}{1 + e^{-\theta^T x}}$$

$$P(y = 1) = p = \sigma(\theta^T x)$$



Some properties,

$$
\begin{aligned}
g'(z) &= \frac{d}{dz}\frac{1}{1+e^{-z}} \\
&= \frac{1}{(1+e^{-z})^2}\left(e^{-z}\right) \\
&= \frac{1}{(1+e^{-z})}\cdot\left(1-\frac{1}{(1+e^{-z})}\right) \\
&= g(z)(1-g(z)).
\end{aligned}
$$

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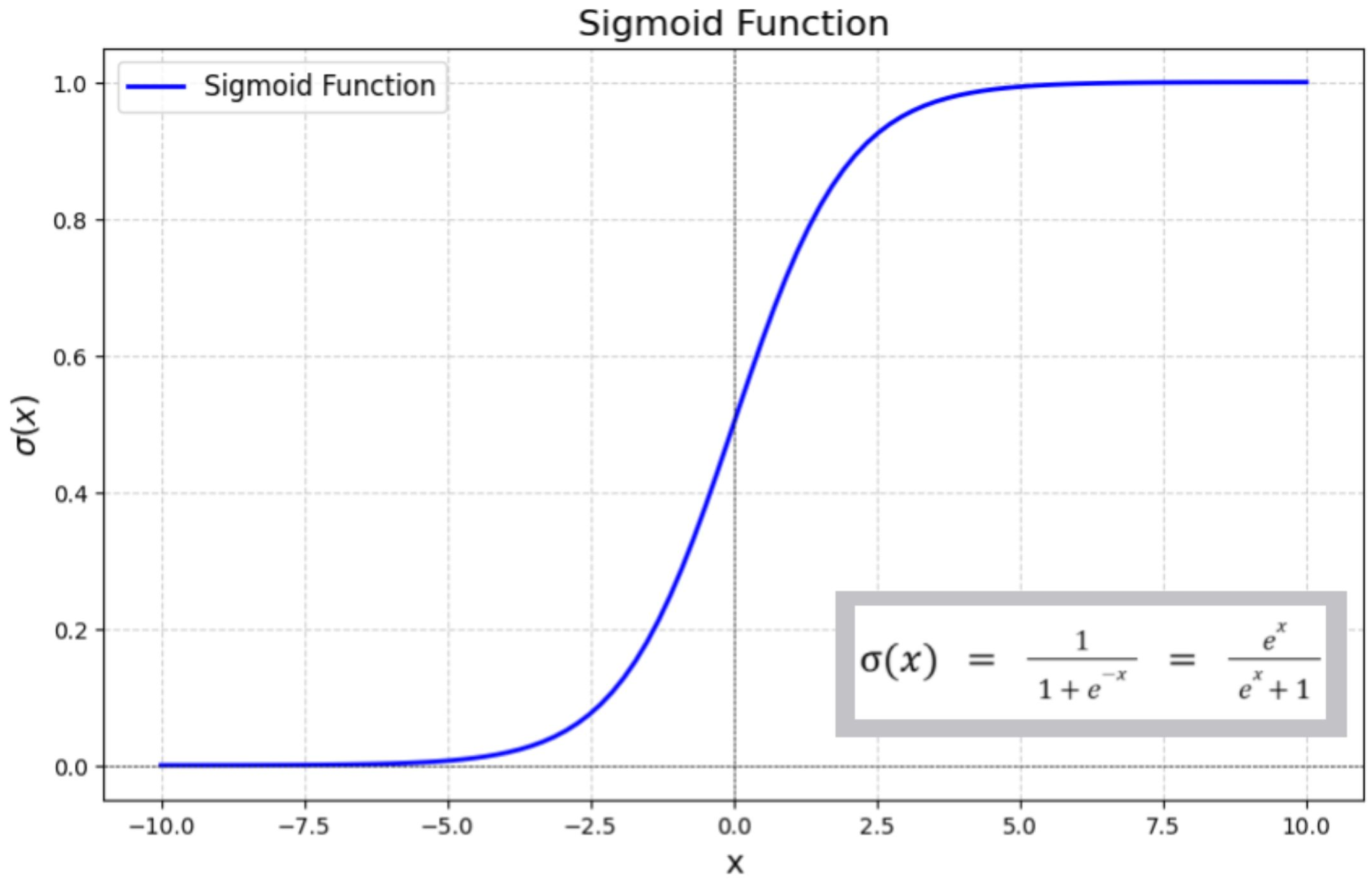
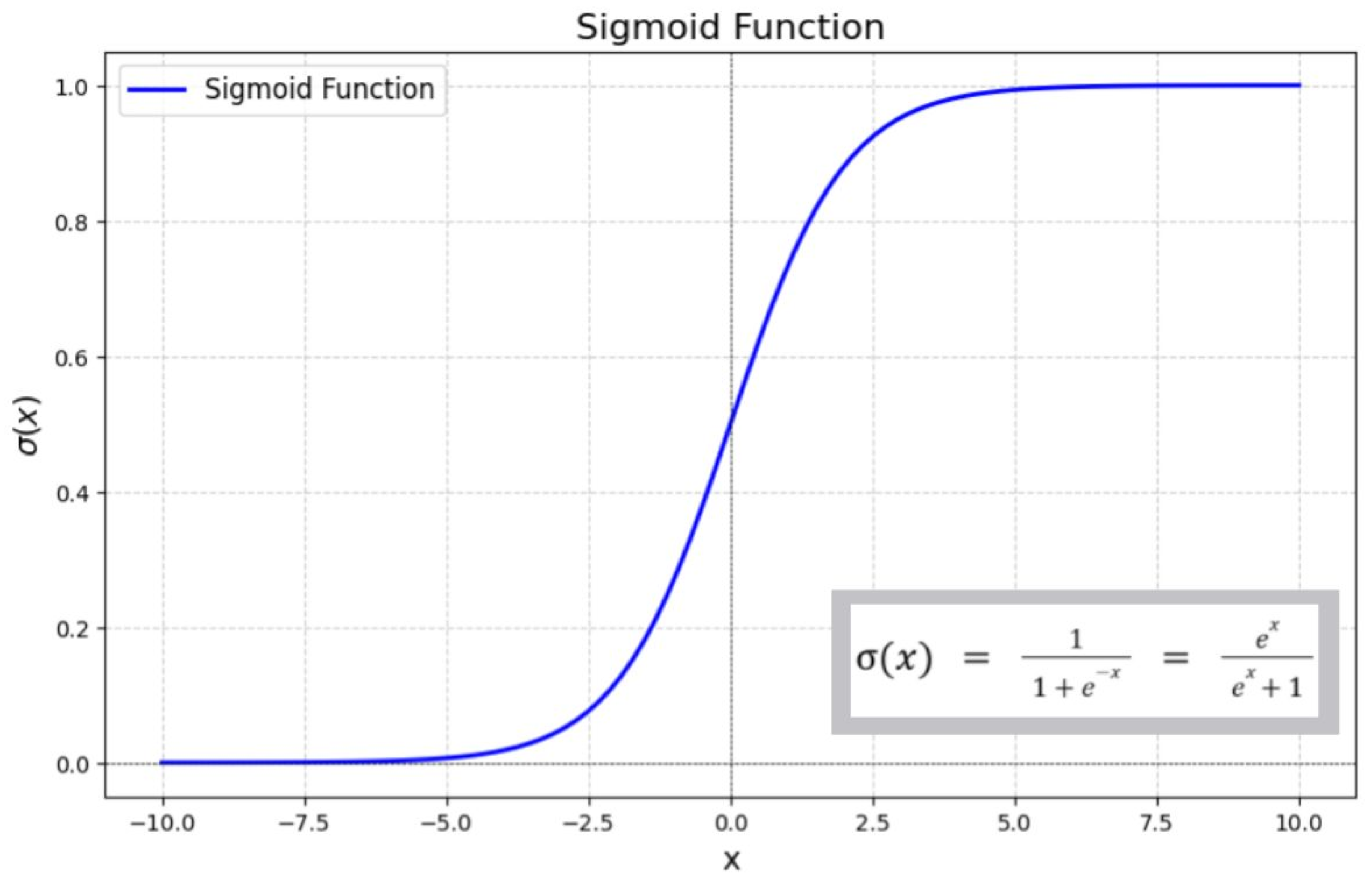


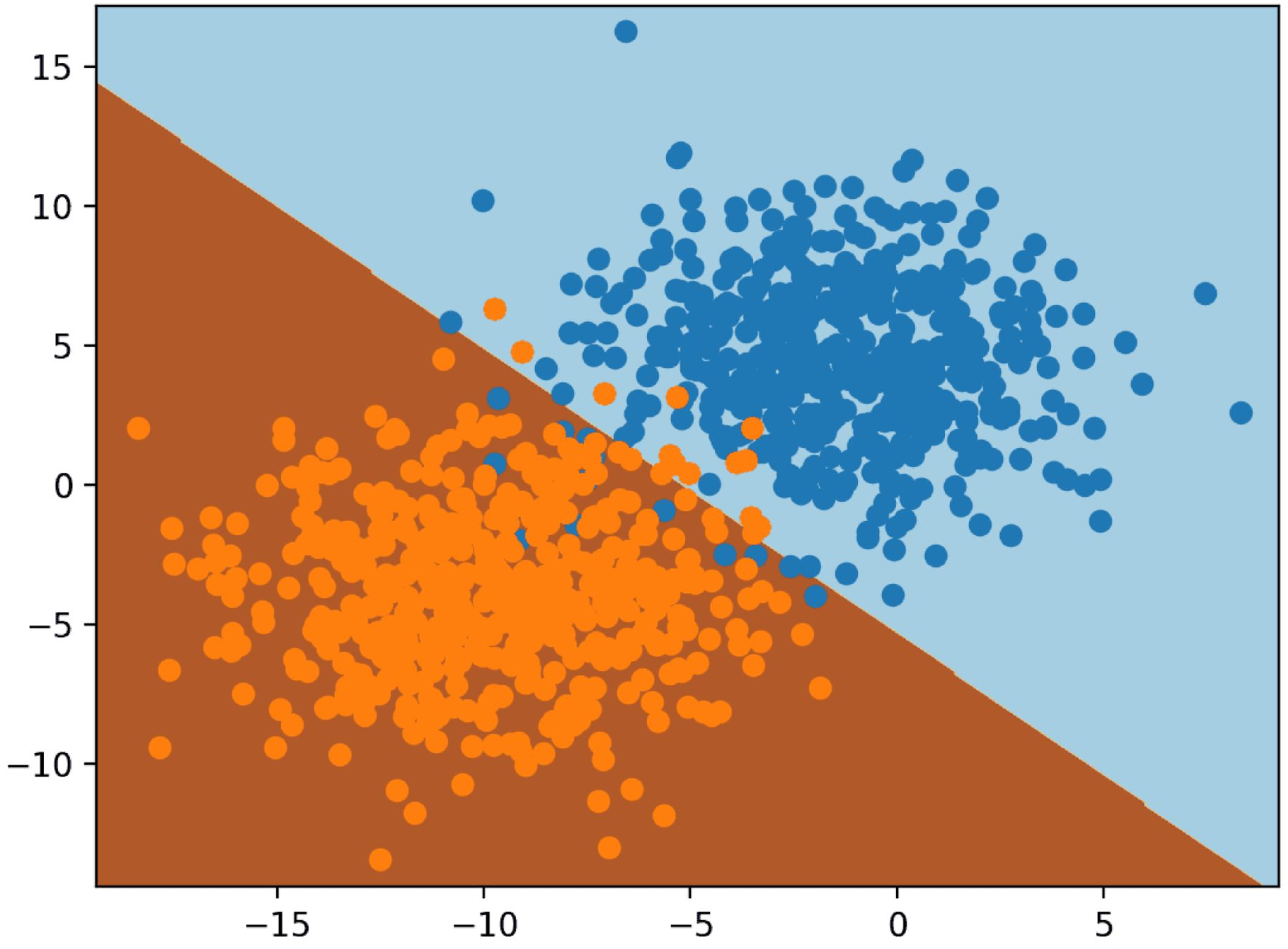
How t

Maxin

$$h_\theta(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}},$$

$$g(z) = \frac{1}{1 + e^{-z}}$$





We assume

$$P(y=1 \mid x;\theta) = h_\theta(x)$$
$$P(y=0 \mid x;\theta) = 1 - h_\theta(x)$$

Or,

$$p(y \mid x;\theta) = (h_\theta(x))^y \,(1-h_\theta(x))^{1-y}$$

Likelihood

$$\begin{aligned} L(\theta) &= p(\vec{y} \mid X;\theta) \\ &= \prod_{i=1}^{n} p(y^{(i)} \mid x^{(i)};\theta) \\ &= \prod_{i=1}^{n} \left(h_\theta(x^{(i)})\right)^{y^{(i)}} \left(1-h_\theta(x^{(i)})\right)^{1-y^{(i)}} \end{aligned}$$

$$\begin{aligned}
L(\theta) &= p(\vec{y} \mid X; \theta) \\
&= \prod_{i=1}^{n} p(y^{(i)} \mid x^{(i)}; \theta) \\
&= \prod_{i=1}^{n} \left(h_\theta(x^{(i)})\right)^{y^{(i)}} \left(1 - h_\theta(x^{(i)})\right)^{1-y^{(i)}}
\end{aligned}$$

$$\begin{aligned}
\ell(\theta) &= \log L(\theta) \\
&= \sum_{i=1}^{n} y^{(i)} \log h(x^{(i)}) + (1 - y^{(i)}) \log(1 - h(x^{(i)}))
\end{aligned}$$

$$\theta_j := \theta_j + \alpha \left(y^{(i)} - h_\theta(x^{(i)})\right) x_j^{(i)}$$

$$\theta_j := \theta_j + \alpha \sum_{i=1}^{m} \left(y^{(i)} - h_\theta(x^{(i)})\right) x_j^{(i)}$$

$$h_\theta(x) = g(\theta^T x)$$

$$g(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

$$\theta_j := \theta_j + \alpha \left(y^{(i)} - h_\theta(x^{(i)})\right) x_j^{(i)}$$